

REPUBLIC OF THE PHILIPPINES · DEPARTMENT OF EDUCATION

Grade 6 Mathematics

Term 2 — Lesson Plans (Lesson Exemplars)

SY 2026–2027 · MATATAG

Ratio & Proportion, Percent, Angles, Transformations, Volume
4 A's format · aligned to the official Budget of Work

Lesson Plans

Six lesson exemplars covering Term 2 competencies

Procedure follows Activity → Analysis → Abstraction → Application.

Lesson 1 · Ratio and Proportion NA 6NA-IIa-1

Subject Matter	Expressing/simplifying ratios; finding the missing term in a proportion
Time	2 sessions (100 min)
Materials	colored counters, recipe cards, map with scale

LEARNING OBJECTIVES

(K) define ratio and proportion and state the cross-products rule; **(S)** express/simplify ratios and solve for a missing term; **(V)** value fairness and balance.

ACTIVITY — “Sago’t Gulaman.” Groups mix “drinks” using 2 cups syrup to 5 cups water and describe the relationship between the amounts.

ANALYSIS — How do we write “2 to 5”? If we double the recipe, how much of each? Does the taste (relationship) change?

ABSTRACTION — A **ratio** compares two quantities (2:5 or $\frac{2}{5}$); simplify like fractions. A **proportion** states two equal ratios; solve a missing term by cross-multiplication: $\frac{2}{5} = \frac{n}{15} \rightarrow 5n = 30 \rightarrow n = 6$.

APPLICATION — “3 notebooks cost ₱75. How much do 8 notebooks cost?” (Answer: $\frac{3}{75} = \frac{8}{n} \rightarrow n = ₱200$.)

FORMATIVE ASSESSMENT / ASSIGNMENT

3-item exit pass. Assignment: find a ratio in your kitchen recipe.

Lesson 2 • Direct and Partitive Proportion NA 6NA-I Ib-2

Subject Matter	Solving problems involving direct and partitive proportion
Time	2 sessions (100 min)
Materials	play money, sharing cards, chart paper

LEARNING OBJECTIVES

(K) distinguish direct from partitive proportion; **(S)** solve real-life proportion problems; **(V)** show fairness in sharing.

ACTIVITY — “Sharing the Harvest.” Divide ₱600 among 3 children in the ratio 1:2:3 using play money.

ANALYSIS — When more workers means more pay, what kind of relationship is it (direct)? When one amount is split into parts (partitive), how do we find each share?

ABSTRACTION — **Direct:** as one quantity increases, the other increases proportionally. **Partitive:** a whole is divided into parts following a ratio. For 1:2:3, total parts = 6; $₱600 \div 6 = ₱100/\text{part} \rightarrow ₱100, ₱200, ₱300$.

APPLICATION — “5 kg of rice costs ₱250. How much do 12 kg cost?” (Answer: ₱50/kg $\rightarrow$ **₱600.**)

FORMATIVE ASSESSMENT / ASSIGNMENT

Pair board work. Assignment: share ₱120 between two siblings in ratio 2:3.

Lesson 3 • Percent, Rate, and Base

NA

6NA-IIc-3

Subject Matter	Relating percent to fractions/decimals; finding percentage, rate, and base
Time	2 sessions (100 min)
Materials	10×10 grids, sale tags, mini Techan's triangle

LEARNING OBJECTIVES

(K) convert among percent, fractions, and decimals; (S) find percentage, rate, or base; (V) value wise spending.

ACTIVITY — “Color the Grid.” On a 10×10 grid learners shade 25 squares and express it as 25%, 0.25, and $\frac{1}{4}$.

ANALYSIS — What does “percent” mean (per hundred)? In “20% of 50 is 10,” which is the rate, base, percentage?

ABSTRACTION — **Percentage = Rate × Base.** Rate is the percent, Base is the whole, Percentage is the part. Example: 15% of 200 = $0.15 \times 200 = 30$. To find rate: 30 is what % of 200 → $30 \div 200 = 15\%$.

APPLICATION — “A class of 40 has 30 present. What percent are present?” (Answer: $30 \div 40 = 75\%$.)

FORMATIVE ASSESSMENT / ASSIGNMENT

3-item quiz card. Assignment: write 3 percents you see in stores as fractions.

Lesson 4 • Percent in Daily Life

NA

6NA-IIId-4

Subject Matter	Discount, commission, sales tax, and simple interest
Time	2 sessions (100 min)
Materials	sale flyers, mock bank slips, calculators (optional)

LEARNING OBJECTIVES

(K) define discount, commission, sales tax, simple interest; (S) solve related percent problems; (V) become a smart consumer and saver.

ACTIVITY — “Sale sa Mall!” Groups compute the sale price of items marked “20% off.”

ANALYSIS — Is the discount added or subtracted? How is commission like a reward for selling? Why does the government collect sales tax?

ABSTRACTION — **Discount** = rate $\times$ original; sale price = original – discount. **Commission** = rate $\times$ sales. **Sales tax** = rate $\times$ price (added). **Simple Interest** $I = P \times R \times T$.

APPLICATION — “A ₱1,500 bag is 20% off. Sale price?” (Discount ₱300 $\rightarrow$ ₱1,200.) “₱5,000 saved at 3% per year for 2 years earns how much interest?” ($I = 5000 \times 0.03 \times 2 =$ ₱300.)

FORMATIVE ASSESSMENT / ASSIGNMENT

Real-flyer task. Assignment: compute interest on ₱2,000 at 5% for 1 year.

Lesson 5 · Angles — Right, Acute, Obtuse

MG**6MG-IE-5**

Subject Matter	Classifying and measuring angles
Time	1–2 sessions (50–100 min)
Materials	protractors, paper fasteners + strips (angle makers), objects in the room

LEARNING OBJECTIVES

(K) classify angles as acute ($<90^\circ$), right ($=90^\circ$), obtuse ($>90^\circ$); **(S)** measure angles with a protractor; **(V)** appreciate accuracy.

ACTIVITY — “*Angle Hunt.*” Learners find and sketch angles around the room (book corner, open door, clock hands).

ANALYSIS — Which angles look “square”? Smaller? Wider? How can we measure exactly?

ABSTRACTION — Demonstrate protractor use (align vertex and base line, read the correct scale). Acute $< 90^\circ$; Right $= 90^\circ$; Obtuse between 90° and 180° ; Straight $= 180^\circ$.

APPLICATION — Measure and classify 4 given angles; draw a 45° , a 90° , and a 120° angle.

FORMATIVE ASSESSMENT / ASSIGNMENT

Classify-5-angles slip. Assignment: find 3 angles at home and classify them.

Lesson 6 • Transformations & Volume MG 6MG-IIIf-6 / 6MG-IIIg-7

Subject Matter	Translation, reflection, rotation; converting volume/capacity units; volume of cubes & rectangular prisms
Time	2 sessions (100 min)
Materials	cut-out shapes, mirror, grid paper, unit cubes, boxes, measuring cups

LEARNING OBJECTIVES

(K) describe the effect of slides, flips, and turns; state $V = l \times w \times h$ and $V = s^3$; **(S)** identify transformations and compute volume; convert $\text{mL} \leftrightarrow \text{L}$ and $\text{cm}^3 \leftrightarrow \text{L}$; **(V)** value spatial care.

ACTIVITY — “Slide, Flip, Turn.” Learners move a cut-out shape on grid paper and name each move; then fill a box with unit cubes and count.

ANALYSIS — Does the shape change size when slid/flipped/turned? How many cubes fit in one layer $\times$ how many layers?

ABSTRACTION — **Translation** = slide, **reflection** = flip (mirror image), **rotation** = turn about a point; size/shape stay the same. **Volume:** rectangular prism $V = l \times w \times h$; cube $V = s^3$. $1 \text{ L} = 1000 \text{ mL} = 1000 \text{ cm}^3$.

APPLICATION — “A box is $8 \text{ cm} \times 5 \text{ cm} \times 4 \text{ cm}$. Find its volume.” ($V = 160 \text{ cm}^3$.) “A cube has edge 3 cm .” ($V = 27 \text{ cm}^3$.)

FORMATIVE ASSESSMENT / ASSIGNMENT

Identify-the-transformation + 2 volume items. Assignment: find the volume of a box at home.